SEPARATED ALGEBRAIC SPACES CHARACTERIZED BY AFFINENESS AND SURJECTIVITY OF FIBRE PRODUCTS OF AFFINES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 04SG: sources/stacks-chow/tags.txt, source lines 6219–6219. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03KD: sources/stacks-chow/tags.txt, source lines 4673–4673. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01JS: sources/stacks-chow/tags.txt, source lines 2025–2025. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01QO: sources/stacks-chow/tags.txt, source lines 2273–2273. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. The following are equivalent*

(1) *X is a separated algebraic space,*
(2) *for $U \to X$, $V \to X$ with U, V affine the fibre product $U \times_X V$ is affine and*
$$\mathcal{O}(U) \otimes_{\mathbf{Z}} \mathcal{O}(V) \longrightarrow \mathcal{O}(U \times_X V)$$
is surjective.

Proof. Using Spaces, Lemma 04SG we see that we may assume $S = \mathrm{Spec}(\mathbf{Z})$. Since $U \times_X V = X \times_{X \times X} (U \times V)$ and since $U \times V$ is affine if U and V are so, we see that (1) implies (2). Assume (2). Choose a scheme W and a surjective étale morphism $W \to X$. Then $W \times W \to X \times X$ is surjective étale. Hence it suffices to show that
$$j : W \times_X W = X \times_{(X \times X)} (W \times W) \to W \times W$$

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

is a closed immersion, see Spaces, Lemma 03KD. If $U \subset W$ and $V \subset W$ are affine opens, then $j^{-1}(U \times V) = U \times_X V$ is affine by assumption and the map $U \times_X V \to U \times V$ is a closed immersion because the corresponding ring map is surjective. Since the affine opens $U \times V$ form an affine open covering of $W \times W$ (Schemes, Lemma 01JS) we conclude by Morphisms, Lemma 01QO. $\square$